

CHE 201 Fall 2019 Final Exam

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TOTAL POINTS

83.5 / 100

QUESTION 1

1 15 pts

1.1 a 10 / 12

- 0 pts Correct
- 2 pts first answer is incorrect
- 2 pts second answer incorrect
- 2 pts third answer is incorrect
- 2 pts fourth answer is incorrect
- 2 pts fifth answer is incorrect
- ✓ - 2 pts sixth answer is incorrect

1.2 b 0 / 3

- 0 pts Correct
- ✓ - 3 pts incorrect
- 2 pts explanation is wrong

QUESTION 2

2 10 pts

2.1 a 3 / 5

- 0 pts Correct
- 2 pts formula is wrong
- 5 pts incorrect
- ✓ - 2 pts temperature is not converted to K

2.2 b 5 / 5

- ✓ - 0 pts Correct
- 5 pts incorrect

QUESTION 3

10 pts

3.1 a 1 / 2

- 0 pts Correct
- 2 pts incorrect

✓ - 1 pts negative sign missing

3.2 b 2 / 2

- ✓ - 0 pts Correct
- 2 pts incorrect

3.3 C 6 / 6

- ✓ - 0 pts Correct
- 1 pts calculation error
- 5 pts Incorrect or no values
- 6 pts Incorrect or no answer
- 2 pts Did not multiply by mass
- 2 pts Wrong value for u
- 3 pts Did not show the calculations for the interpolate values
- 2 pts T should be in Kelvin

QUESTION 4

32 pts

4.1 a 2 / 2

- ✓ - 0 pts Correct
- 1 pts boundary is wrong

4.2 b 3 / 3

- ✓ - 0 pts Correct
- 1 pts closed system missing
- 1 pts constant pressure missing
- 1 pts $\Delta KE = \Delta PE = 0$ missing

4.3 C 4 / 4

- ✓ - 0 pts Correct
- 2 pts Wrong simplified energy balance
- 2 pts Wrong simplified entropy balance
- 1 pts Need to simplify the entropy balance

4.4 d 8 / 8

✓ - 0 pts Correct

- 1 pts unit conversion for work
- 2 pts v_2 is wrong
- 1 pts P is wrong
- 1 pts Work is not multiplied with mass
- 4 pts Calculation of Q is wrong
- 4 pts Calculation of W is wrong
- 8 pts Wrong answer
- 2 pts u_2 is wrong
- 1 pts U_1 is wrong
- 1 pts v_1 is wrong
- 1 pts Calculation error
- 2 pts States 1 and 2 are inverted

4.5 e 5 / 5

✓ - 0 pts Correct

- 1 pts need to multiply by mass
- 5 pts Wrong or no answer
- 1 pts Should use Table A-2 or interpolate Table A-3
- 2 pts s_2 is wrong
- 2 pts s_1 is wrong
- 1 pts Calculation error

4.6 f 4 / 5

- 0 pts Correct
- ✓ - 1 pts T is wrong should be boundary T
- 2 pts Did not state if process is reversible, irreversible or impossible
- 3 pts Wrong solution
- 1 pts Process is impossible
- 1 pts Calculation or equation error
- 5 pts No answer

4.7 g 3 / 5

- 0 pts Correct
- 1 pts x axis not labeled
- ✓ - 2 pts T-s graph is wrong or absent
- 1 pts v values are wrong or absent
- 1 pts s value is wrong or absent
- 2 pts P-v graphs is wrong

- 1 pts y axis not labeled
- 1 pts Wrong or missing units

QUESTION 5

5 33 pts

5.1 a 3 / 4

- 0 pts Correct
- 4 pts Incorrect
- ✓ - 1 pts Did not represent Work
- 1 pts Wrong system boundary
- 1 pts Missing one outlet

5.2 b 3 / 5

- 0 pts Correct
- 1 pts steady state missing
- 2 pts control volume missing
- ✓ - 1 pts $\Delta PE = 0$ missing
- ✓ - 1 pts Cannot assume $\Delta KE = 0$

5.3 c 6 / 6

- ✓ - 0 pts Correct
- 2 pts entropy balance is wrong
- 2 pts energy balance is wrong
- 2 pts Q, W, m is not written in rate format
- 1 pts Need to simplify entropy balance

5.4 d 8.5 / 10

- 0 pts Correct
- 1 pts calculation error
- ✓ - 1.5 pts unit conversion error
- 2 pts h_1 is wrong
- 2 pts h_2 is wrong
- 2 pts h_3 is wrong
- 8 pts Incomplete answer
- 1 pts Answer should be in MW
- 1 pts m_1 is wrong
- 2 pts m_2 and m_3 are wrong
- 1 pts incomplete calculation

5.5 e 7 / 8

- 0 pts Correct

- **8 pts** incorrect or no answer
- **1 pts** calculation error
- **2 pts** s2 is wrong or absent
- **2 pts** s3 is wrong or absent
- **2 pts** T is wrong
- **4 pts** Incorrect equation or no equation
- **2 pts** m2 and m3 are wrong or absent
- ✓ - **1 pts** unit conversion error
- **2 pts** s1 is wrong

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Total points: 100

Please fill in your answers in the space supplied

Show all of your calculations, conversions and work to receive full credit.

QUESTION 1 (15 pts)

True or False (2 pts each)True The upstream and downstream enthalpies of flow through an insulated valve is equal.False Entropy, S, is path-dependent.False Helmholtz energy is an intensive property.True A closed system can experience an increase in entropy only when there is heat transfer to the system.True The change in Gibbs free energy (dg) is equal to zero for a phase change from a saturated liquid to a saturated vapor.False The second Carnot corollary states that reversible power cycles operating between the same two thermal reservoirs have the same thermal efficiency.**Fill in the blank**

(3 pts) A gas goes from State A to State B in a reversible, adiabatic process. It then goes from State B back to State A by a different pathway that is irreversible and not adiabatic. The entropy change, (ΔS) for the irreversible pathway is

greater than (greater than/equal to/ less than) the entropy change, (ΔS) for the reversible pathway. Explain.

$$\Delta S = \int_1^2 \frac{dq}{T} + \int_2^1 \frac{dq}{T}$$

$$\Delta S_1 = 0 \text{ reversible not adiabatic}$$

$$\Delta S = \int_1^2 \frac{dq}{T} + \int_2^1 \frac{dq}{T} \leftarrow \text{greater than zero}$$

QUESTION 2 (10 pts)

- a) (5 pts) A steam engine operates with a theoretical maximum efficiency of %40 and a minimum temperature of 100°C. Calculate temperature of the engine to maintain maximum efficiency.

$$\eta = 1 - \frac{T_c}{T_H}$$

$$\frac{T_c}{T_H} = 1 - 0.40$$

$$T_H = \frac{100}{0.60} = 166.67^\circ\text{C}$$

- b) (5 pts) The Reynolds number, Re, is a dimensionless number that measures the ratio of inertial forces to viscous forces in fluid flow;

$$Re = \frac{\rho V L}{\mu}$$

In this equation, ρ is the density, V is the velocity, L is a characteristic length, and μ is the viscosity. What are the dimensions of μ ?

Show all your work for full credit.

$$\rho = \frac{\text{kg}}{\text{m}^3} \quad V = \frac{\text{m}}{\text{s}} \quad L = \text{m}$$

$$Re = \frac{\rho V L}{\mu}$$

$$\mu \cdot Re = \rho V L$$

$$\mu = \frac{\rho V L}{Re}$$

$$= \frac{\text{kg} \cdot \text{m} \cdot \text{m}}{\text{m}^3 \cdot \text{s}} = \frac{\text{kg}}{\text{m} \cdot \text{s}}$$

QUESTION 3 (10 pts)

- a) (2 pts) Write the specific Maxwell relation that corresponds to the thermodynamic function:

$$u(s, v): \left(\frac{\partial T}{\partial v} \right)_s = \left(\frac{\partial P}{\partial s} \right)_v$$

$$du = Tds - pdv$$

$$u = \left(\frac{\partial T}{\partial v} \right)_s = \left(\frac{\partial P}{\partial s} \right)_v$$

- b) (2 pts) Write the two, second mixed partial derivatives for the exact differential:

$$dh = Tds + vdP : \frac{dh}{dsdP} = \left(\frac{\partial T}{\partial P} \right)_s = \left(\frac{\partial v}{\partial s} \right)_P$$

$$\frac{dh}{dsdP} = \left(\frac{\partial T}{\partial P} \right)_s = \left(\frac{\partial v}{\partial s} \right)_P$$

- b) (6 pts) A 3 kg mass of saturated liquid water has a pressure of 12.0 bar. What is the Helmholtz energy, in kJ?

Show all your work for full credit

$$\psi = u - Ts$$

$$\psi = 794.45 \frac{\text{kJ}}{\text{kg}} - \left(460.26 \text{ K} \times 2.209 \frac{\text{kJ}}{\text{kg} \cdot \text{K}} \right)$$

$$= -667.097 \text{ kJ}$$

$$\frac{10-12}{12-15} = \frac{179.9-15}{x-198.3}$$

$$(10-12)(x-198.3) = (12-15)(179.9-x)$$

$$\frac{15-12}{12-10} = \frac{198.3-x}{x-179.9}$$

$$3(x-179.9) = 2(198.3-x)$$

$$3x - 539.7 = 396.6 - 2x$$

$$3x + 2x = 396.6 + 539.7$$

$$5x = 936.3$$

$$x = 187.26^\circ\text{C}$$

$$\frac{15-12}{12-10} = \frac{843.6-x}{x-761.68}$$

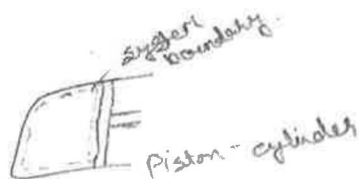
$$u = 794.45 \frac{\text{kJ}}{\text{kg}}$$

$$s = \frac{15-12}{12-10} = \frac{12.3150-x}{x-32.1387}$$

$$s = 2.209 \frac{\text{kJ}}{\text{kg} \cdot \text{K}}$$

3 kg of water, initially a saturated liquid at 150°C is contained piston-cylinder device. The water undergoes a process. Heat is transferred between the piston-cylinder and a 25°C room until the water becomes a state where $x=0.8$. The piston moves freely and maintains constant pressure throughout the process. Kinetic and potential energy can be ignored.

- (2 pts) Draw a schematic of the system and label the system boundary.
- (3 pts) State the engineering model
- (4 pts) Write the simplified energy balance and simplified entropy balance.
- (8 pts) Determine the work in **kJ**, and the heat transferred in **kJ**.
- (5 pts) Determine the change in entropy of the water during the process, **ΔS , in kJ/K**
- (5 pts) Determine the entropy generated during the process, **σ , in kJ/K**. State if this process is reversible, irreversible or impossible.
- (5 pts) Carefully sketch the process on P-v and T-s diagram.



- Piston cylinder
- closed system
- constant pressure
- $\Delta KE = \Delta PE = 0$
- piston moves freely
- Heat transferred

- Energy balance $\Delta u + \Delta KE + \Delta PE = Q - W$
 $\Delta u = Q - W$

- Entropy balance $\Delta S = \int_1^2 \left(\frac{\delta Q}{T} \right)_b + \sigma$
 $S_2 - S_1 = \frac{Q_{in}}{T} + \sigma$

$$W = \int_{V_1}^{V_2} P dV = P \int_{V_1}^{V_2} dV = P(V_2 - V_1)$$

State 1 = Saturated liquid and $T = 150^\circ\text{C}$

$$v_1 = 1.0905 \times 10^{-3} \text{ m}^3/\text{kg} = 1.0905 \times 10^{-3} \times 3 = 0.00327 \text{ m}^3$$

$$S_1 = 1.8418 \frac{\text{kJ}}{\text{kg} \cdot \text{K}} \times 3 \text{ kg} = 5.525 \frac{\text{kJ}}{\text{K}}$$

$$u_1 = 631.68 \frac{\text{kJ}}{\text{kg}} \times 3 = 1895.04 \text{ kJ}$$

State 2:- $T = 150^\circ\text{C}$, $x = 0.8$

$$V_2 = V_F + x(V_g - V_F) \\ = (1.0905 \times 10^{-3} + 0.8(0.3928 - 1.0905 \times 10^{-3})) \times 3 \text{ kg} \\ = 0.943 \text{ m}^3$$

$$u_2 = 150^\circ\text{C}, x = 0.8$$

$$u_2 = 631.68 + 0.8(2559.5 - 631.68) \\ = 2173.94 \frac{\text{kJ}}{\text{kg}} \times 3 \text{ kg} = 6521.808 \text{ kJ}$$

$$S_2 = 1.8418 + 0.8(6.8379 - 1.8418) \\ = 5.83868 \times 3 = 17.516 \text{ kJ/K}$$

$$W = P(V_2 - V_1) = 4.758 \times 10^5 (0.943 \text{ m}^3 - 0.00327 \text{ m}^3) \\ = \frac{447123.534 \text{ Nm}}{\text{Nm} / 1000} \frac{\text{J}}{\text{Nm}} \\ = 447.123 \text{ kJ (Work done by system)}$$

$$\Delta u = Q - W$$

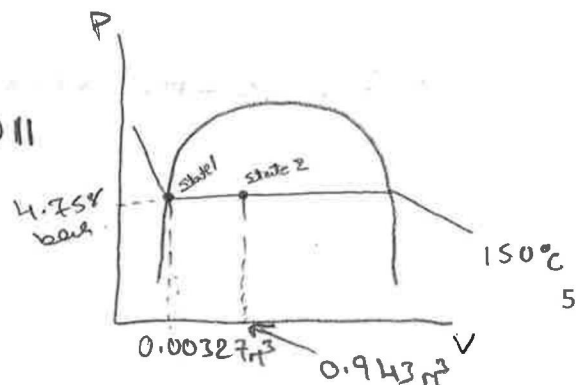
$$\therefore Q = \Delta u + W \\ = (u_2 - u_1) + W = (6521.808 - 1895.04) + 447.123 \\ = 5073.89 \text{ kJ}$$

$$\Delta S = S_2 - S_1 = 17.516 \frac{\text{kJ}}{\text{K}} - 5.525 = 11.991 \frac{\text{kJ}}{\text{K}}$$

$$\Delta S = \frac{Q_{\text{net}}}{T} + \sigma$$

$$\sigma = 11.991 - \frac{5073.89}{423} = -0.004011$$

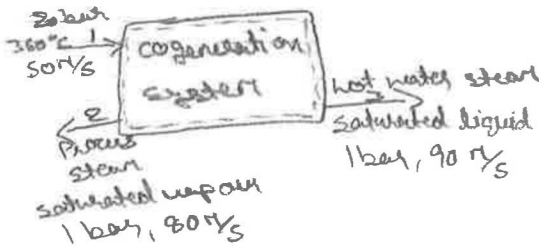
so this process is impossible



QUESTION 5: (33 pts)

A cogeneration system operates at steady state. Water vapor enters the system at 20 bar, 360°C, velocity of 50 m/s and mass flow rate of 1.5 kg/s. Power is developed by the system at a rate of 2.5 MW. There are two outlets for the system: process steam and hot water. Process steam leaves as saturated vapor at 1 bar, 80 m/s, and hot water leaves as saturated liquid at 1 bar, 90 m/s. Mass flow rates of process steam and hot water are equal. Neglect any changes in potential energy.

- (4pts) Draw a schematic of the system and label the system boundary.
- (5pts) State the engineering model.
- (6pts) Write simplified balances for mass, energy and entropy.
- (10pts) Calculate the rate of heat transfer, in MW, between the system and its surroundings.
- (8pts) Calculate the entropy production for the process, in kJ/K-s.



$$W = 2.5 \text{ MW}$$

- steady state (cogeneration system)
- control volume
- $\Delta KE = 0$
- One inlet, two outlet

Mass balance

$$\frac{dm_{cv}}{dt} = \dot{m}_i - \dot{m}_e$$

$$0 = \dot{m}_1 - (\dot{m}_2 + \dot{m}_3)$$

$$\therefore \dot{m}_1 = \dot{m}_2 + \dot{m}_3$$

$$\therefore 1.5 = \dot{m}_2 + \dot{m}_3$$

$$\therefore \dot{m}_2 = 0.75 \quad \dot{m}_3 = 0.75$$

Energy balance

$$\frac{dE_{cv}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \sum \dot{m}_i \left(h_i + \frac{V_i^2}{2} + gz_i \right) - \sum \dot{m}_e \left(h_e + \frac{V_e^2}{2} + gz_e \right)$$

$$0 = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}_1 \left(h_1 + \frac{V_1^2}{2} \right) - \left(\dot{m}_2 \left(h_2 + \frac{V_2^2}{2} \right) + \dot{m}_3 \left(h_3 + \frac{V_3^2}{2} \right) \right)$$

Entropy balance

$$\frac{dS_{cv}}{dt} = \sum \frac{\dot{Q}_i}{T_i} + \sum \dot{m}_i s_i - \sum \dot{m}_e s_e + \dot{\sigma}_{cv}$$

$$0 = \frac{\dot{Q}}{T} + \dot{m}_1 s_1 - (\dot{m}_2 s_2 + \dot{m}_3 s_3) + \dot{\sigma}_{cv}$$

$$\dot{Q}_{cv} = \dot{W}_{cv} - \dot{m}_1 \left(h_1 + \frac{V_1^2}{2} \right) + \dot{m}_2 \left(h_2 + \frac{V_2^2}{2} \right) + \dot{m}_3 \left(h_3 + \frac{V_3^2}{2} \right)$$

$$\text{State 1} \rightarrow h_1 = 3159.3 \frac{\text{kJ}}{\text{kg}}$$

$$\text{State 2} \rightarrow h_2 = 2675.5 \frac{\text{kJ}}{\text{kg}}$$

$$\text{State 3} \rightarrow h_3 = 417.46 \frac{\text{kJ}}{\text{kg}}$$

$$\begin{aligned} 1 \text{ MW} &= 10^9 \text{ W} \\ 1 \text{ kW} &= 10^3 \text{ W} \\ \dot{W}_{cv} &= 2.5 \times 10^6 \text{ kW} \end{aligned}$$

$$\begin{aligned} \dot{Q}_{cv} &= 2.5 \times 10^6 \text{ kW} - 1.5 \left(3159.3 + \frac{50^2}{2} \right) + 0.75 \left(2675.5 + \frac{80^2}{2} \right) + \\ &\quad 0.75 \left(417.46 + \frac{90^2}{2} \right) \end{aligned}$$

$$= \frac{2501143.27 \text{ kW}}{1 \text{ kW}} \left| \frac{10^3 \text{ W}}{1 \text{ kW}} \right| \left| \frac{1 \text{ MW}}{10^3 \text{ W}} \right|$$

$$= 2.501 \text{ MW}$$

$$0 = \frac{\dot{Q}}{T} + \dot{m}_1 s_1 - (\dot{m}_2 s_2 + \dot{m}_3 s_3) + \dot{S}_{cv}$$

$$\dot{S}_{cv} = -\frac{\dot{Q}}{T} - \dot{m}_1 s_1 + \dot{m}_2 s_2 + \dot{m}_3 s_3$$

$$s_1 = 6.9917 \frac{\text{kJ}}{\text{kg} \cdot \text{K}}$$

$$s_2 = 7.3594 \frac{\text{kJ}}{\text{kg} \cdot \text{K}}$$

$$s_3 = 1.3026 \frac{\text{kJ}}{\text{kg} \cdot \text{K}}$$

$$\begin{aligned} \dot{S}_{cv} &= -\frac{2501143.27}{633} - 1.5(6.9917) + 0.75(7.3594) \\ &\quad + 0.75(1.3026) \end{aligned}$$

$$= -3955.24 \frac{\text{kJ}}{\text{K} \cdot \text{s}}$$

$$= -3955.24 \frac{\text{kJ}}{\text{K} \cdot \text{s}}$$

